KAZE-SAR: SAR Image Registration Using KAZE Detector and Modified SURF Descriptor for Tackling Speckle Noise

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***Abstract*— Synthetic aperture radar (SAR) image registration is still a challenging task in remote sensing. The scale-invariant feature transform (SIFT) method and its extensions are most widely used feature detectors in SAR images. However, due to the presence of nonlinear speckle noise, some wrong features are chosen which directly influence the feature matching process. In this article, for the first time, the KAZE algorithm with a modified version of speeded-up robust features (SURF) descriptor is used for SAR image registration. It uses discrete stochastic second order nonlinear partial differential equations (PDEs) to model the SAR image's edge structure. KAZE uses nonlinear diffusion filtering to build up the scale levels of the SIFT descriptor. It preserves the edges while simultaneously decreasing the speckle noise in smoothing the image. Therefore, it captures the exact location of features in down-sampling and filtering stages of SIFT. Experimental results show that our proposed method has increased the localization accuracy and distinctiveness of feature extraction through the use of KAZE detector. Our method is also able to handle challenging cases in different image sources, different view angles, different image times, complex affine mapping, presence of blurring, and noise.**

***Index Terms*— Feature descriptor, feature detector, feature extraction, KAZE, remote sensing, speckle noise, synthetic aperture radar (SAR) image registration.**

I. Introduction

THE synthetic aperture radar (SAR) image is a microwave radar image in remote sensing. Contrary to optical images, it can work in different weather and illumination conditions. SAR images are usually acquired at different times, from different sensors, and from different viewpoints [1], [2].

These images have different translation, rotations, shear, scale, and illumination with respect to each other. Therefore, concatenating these images to create a panorama image from these different images is called image registration [3], [4].

SAR image registration has been widely used in remote sensing. Many new methods have been proposed to solve this problem by using a low-rank matrix [5], sparse theory [6], guided locality preserving feature matching (GLPM) [7], Delaunay-triangulation-based [8], OS-flow [9], Tsallis entropy [10], and arborescence network matching [11] theories.

Scale-invariant feature transform (SIFT) [12] is one of the most commonly used methods of SAR image registration [13]–[16]. The SIFT method is sensitive to speckle noise and directly influences its performance in SAR image registration. Therefore, many extensions of this method such as principal component analysis (PCA)-SIFT [17], modified SIFT [18], multilevel SIFT [19], colored SIFT (C-SIFT) [20], N-dimensional images SIFT (N-SIFT) [21], scale-invariant feature detector with error resilience (SIFER) [22], bilateral filter SIFT (BF-SIFT) [23], improved SIFT [24], SAR-SIFT [25], uniform SIFT [26], and affine SIFT (A-SIFT) [27] have been proposed. In spite of the extensive efforts in this field, SIFT and its extensions are still sensitive to nonlinear speckle noise in SAR images.

Similar efforts have been made in computer vision to find a proper descriptor for image registration [28]–[31]. KAZE [32] is one of the powerful detectors in image processing [2], [33]–[35] which has not been used in SAR image registration. The main strength of this algorithm is its ability to work in noisy conditions. Demchev *et al.* [36] have used advanced KAZE (A-KAZE) for object tracking and it is concentrated in the specific application of sea ice. Liu and Xiao [37] have used KAZE with a binary robust independent elementary features (BRIEF) descriptor for global remote sensing applications. Their descriptor is different from ours and they have not concentrated on speckle noise; therefore, their method is not optimized for SAR image registration.

In this article, the KAZE algorithm, as an extension of the SIFT algorithm, is used for SAR image registration. Our method also has a few distinguishing features. It defines the exact places of SAR image edges in the presence of speckle noise in different pyramid layers of SIFT algorithms. Therefore, it is able to handle speckle noise much better than the SIFT algorithm. It usually finds many more features with respect to speeded-up robust features (SURF) in the same SAR image. It also outperforms some state-of-the-art methods including SURF, maximally stable extremal regions (MSER),

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and binary robust invariant scalable keypoints (BRISK) methods. It can handle complex affine mapping between the sensed images and with reference one including translation, rotation, scale, shear, and transformations simultaneously.

Our method has a few innovations as follows.

1) For the first time, the KAZE detector with the modified SURF descriptor is used for SAR image registration.
2) It uses the discrete stochastic second-order nonlinear partial differential equations (PDEs) to model the SAR image's edge structure.
3) It uses additive operator splitting (AOS) techniques to solve the PDE of the image intensity model. Therefore, an idea and innovation to solve PDE faster are used here to increase the accuracy of SAR image registration.
4) It uses the modified version of the SURF algorithm for fitting nonlinear scale-space model, which is highly matched with the KAZE detector for SAR image description.

This article is organized as follows. Section II describes our preprocessing nonlinear diffusion filtering method. Then the KAZE algorithm is clarified and experimental results are shown in Section III. Section IV concludes this article and delineates the future works.

II. Our Proposed Method

Our proposed method has three steps. At first, KAZE detector is used to extract features. Then, the modified SURF descriptor will be used. Finally, feature matching by random sample consensus (RANSAC) will be carried out.

A. KAZE Detector

The SIFT algorithm has a few steps [12]. At first, it constructs scale spaces. Then, it takes the difference of Gaussians (DoG) and locates their extrema. Hence, it fits a mesh to the surface of the image and calculates the feature point in subpixel. Then it filters edges and eliminates low contrast points. After that, it assigns orientations to each keypoint. Finally, it builds the keypoint descriptor.

Although SIFT is a powerful detector in optical images, it encounters problems in SAR images due to the presence of speckle noise. In fact, SIFT is a gradient-based detector and is highly sensitive to the true place of image edge. Speckle noise causes the location of edge to be inaccurately detected. This error that is propagated in other steps of SIFT and feature matching will have errors. Therefore, there is an essential need to minimize noise while simultaneously preserving edges. This is exactly the work which is done by nonlinear diffusion filtering.

If the evolution of the luminance of an image through increasing scale levels in the SIFT approach is modeled by the divergence of a certain flow function that controls the diffusion process, it would be the milestone of nonlinear filtering. This diffusion process is described by nonlinear PDEs shown in the following equation:

$$U_t = \frac{\partial U}{\partial t} = \operatorname{div}(c(x, y, t) \cdot \nabla U) = c(x, y, t)\Delta U + \nabla c \cdot \nabla U. \tag{1}$$

In (1), U is luminance of an image and time t is the scale parameter and ∇, Δ, and div show the Gradient, Laplacian, and Divergence operations, respectively. The conductivity function c in the diffusion equation allows the diffusion process to be adaptive to the local image structure. In traditional Gaussian filtering, c is a fixed function.

It is proper to have variable conductance diffusion and adjust c such that it has large values at the background of an image and small values at the location of edges [see (2)]. It allows us to control the diffusion at each scale level and preserve image edges from smoothing while smoothing the background and noise. In the following equation, ∇u_σ is the gradient of a smoothed Gaussian version (g) of the original image I:

$$c(x, y, t) = g(\|\nabla u_\sigma(x, y, t)\|). \tag{2}$$

The main problem is that it is not clear where the image edges are located. Perona and Malik [38] proposed that the conductivity function should be a nonlinear continuous monotically decreasing function of Gradient image [see (2)]. In this article, the function of (3) is chosen. In this equation, k is the contrast factor that controls the level of diffusion and is computed empirically (based on [32]) as the 70% percentile of the Gradient histogram of Gaussian smoothed input image. The conductivity function equation (3) promotes wide regions over smaller ones

$$g(\nabla u) = \frac{1}{1 + \left(\frac{\|\nabla u\|}{K}\right)^2}. \tag{3}$$

The result of this smoothing strategy is shown in the SAR image in Fig. 1. It is visible that consequent filtering of the Gaussian filter smooths both edges and background while diffusion filtering preserves the edges while smoothing background and speckle noise.

As the image is a discrete signal, it is necessary to discretize the parabolic PDE of (1). The result of discretization is shown in (4). It is worthy to note that this PDE is stochastic because the exact location of the edge is not defined before and it can be anywhere in the image. PDEs can be solved by the three methods of finite difference, finite volume, and finite element. The finite difference method uses Taylor series to solve equations and is chosen in our approach.

To solve discrete PDE with the finite difference method, there are three methods, which are called explicit, implicit, and Crank–Nicolson. Explicit methods are fast but sometimes numerically unstable. Implicit methods are slow but always numerically stable. Crank–Nicolson is numerically stable but its speed is lower than explicit and better than implicit methods.

To discretize the summands of the right-hand side of (4), only one index for pixel numbering is chosen. Therefore, the entire image of size $N_1 \times N_2$ is represented as a vector size $N := N_1 \dots N_2$. In this vector-matrix notation, the 2-D explicit strategy is as shown in (5)

$$\partial_t u = \sum_{l=1}^{m} \partial_{x_l} \left[g\left(|\nabla u_\sigma|^2\right) \partial_{x_l} u \right] \tag{4}$$

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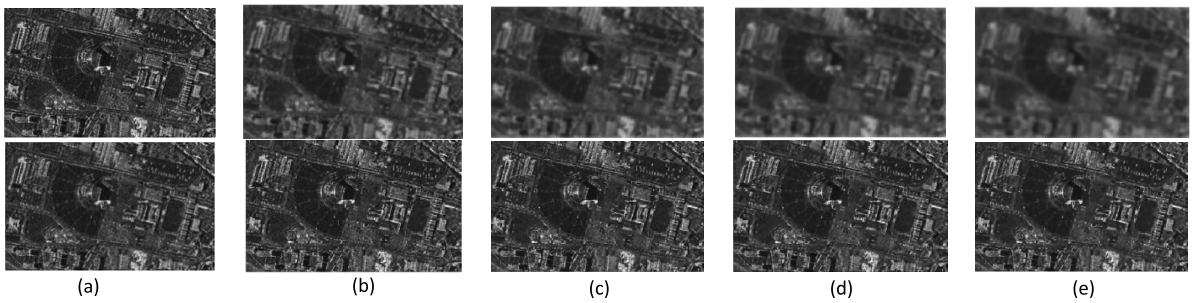


Fig. 1. Consequent exertion of Gaussian (row 1) and nonliner diffusion filtering (row 2) to an SAR image with speckle noise. (a) Original Image. (b) Iteration 1. (c) Iteration 2. (d) Iteration 3. (e) Iteration 4.

$$u^{k+1} = \left[I + \tau \sum_{l=1}^{m} A_l\left(u^k\right) \right] u^k \tag{5}$$

$$u^{k+1} = \left[I - \tau \sum_{l=1}^{m} A_l\left(u^k\right) \right]^{-1} u^k. \tag{6}$$

Weickert *et al.* [39] have proposed the semi-implicit version of the PDE of (4)–(6). The semiexplicit strategy does not directly represent u^{k+1} based on u^k. However, u^{k+1} can be obtained by solving the system of linear equations.

Alcantarilla *et al.* [32] have proposed AOS as a new alternative to solve (4) based on a new semi-implicit strategy. They have changed the place of sigma operation (Σ) of (6) to the new place of (7). Thus, this strategy approximates (6) and solves PDE simpler and faster

$$u^{k+1} = \frac{1}{m} \sum_{l=1}^{m} \left[I - m\tau A_l\left(u^k\right) \right]^{-1} u^k. \tag{7}$$

In other words, a simpler and faster innovation for the solution of second-order PDE is suggested and used to solve PDEs. Then, this innovation in PDE has caused a reduction in speckle noise while preserving edges in SAR images and a numerically stable result.

This strategy is used in the SIFT method and evolving it to the KAZE method improves its localization accuracy and distinctiveness [33]. In fact, in KAZE, building the scale space is similar to SIFT. Then AOS will be used to solve PDE of the filtered image. The DoG step of SIFT is replaced by nonlinear diffusion filtering in KAZE and the other steps of SIFT will remain intact. Then, a set of keypoints are detected which exhibit a maximum of scale-normalized determinants of Hessian response at multiple scale levels.

B. Modified SURF Descriptor

The dominant orientation for a local neighborhood centered around the keypoint is estimated similar to the SURF descriptor [40] using a circular area of radius $6\sigma_i$ and a sampling step σ_i (σ = scale of detected keypoint). The dominant orientation is obtained by summing the responses in a sliding circle covering an angle of $\pi/3$. Then the orientation of the longest vector will be selected. A modified version of the SURF descriptor (MSURF) [41] will be used for building the feature descriptor. A rectangular grid of size $24\sigma_i \times 24\sigma_i$ is considered. Then the first order derivatives L_x and L_y of size σ_i are computed for each keypoint. After that, the grid is divided into overlapping 4×4 subregions of size $9\sigma_i \times 9\sigma_i$. The derivative responses are weighted by a Gaussian kernel centered on the subregion center and each subregion is summed into a descriptor vector ($\upsilon = \{\sum L_x, \sum |L_x|, \sum L_y, \sum |L_y|\}$).

Doing this for all the 4×4 subregions yields a feature descriptor of length $4 \times 4 \times 4 = 64$ dimensions instead of 128 of the SIFT so it is faster than SIFT. Finally, the feature vector is normalized into a unit vector to gain robustness against illumination variations.

The binary descriptors like BRIEF [42], BRISK [43], and oriented FAST and rotated BRIEF (ORB) [44] are faster than SURF descriptor, but they are highly sensitive to absolute pixel values. Therefore, they are not a good choice for noisy image matching, especially the SAR image that has nonlinear speckle noise.

Feature detector directly finds the location of features. If the places of the features are selected wrongly and it is influenced by speckle noise it directly influences on the other next steps of image registration and the error will be propagated. KAZE is selected for this step. However, the descriptor, defines the way that we interpret the detector. If our description is sensitive to the absolute value of pixels, it directly accepts effect of noise. Therefore, a good selection of description is also important. Our modified SURF descriptor is faster than SIFT, while its specific weighted Gaussian kernel enables it to handle noise much better.

C. Feature Matching

After describing the features of two SAR images, the RANSAC algorithm will be used to match the corresponding features of two images. Then, the parameters of the homographic matrix will be estimated based on the affine transformation. As the number of equations (matched points) are usually higher than the unknowns (parameters of the matrix), an over-determined matrix will be shaped. This matrix will be solved by singular value decomposition (SVD) and the right singular value of this matrix will be selected as the final homographic matrix coefficients. Then a simple averaging is considered as our image stitching strategy.

It is also noteworthy that the affine transformation is just an option and can be replaced with any other image matching techniques similar to projective transformation, sparse [6], and guided local features [7]. However, as in SAR image

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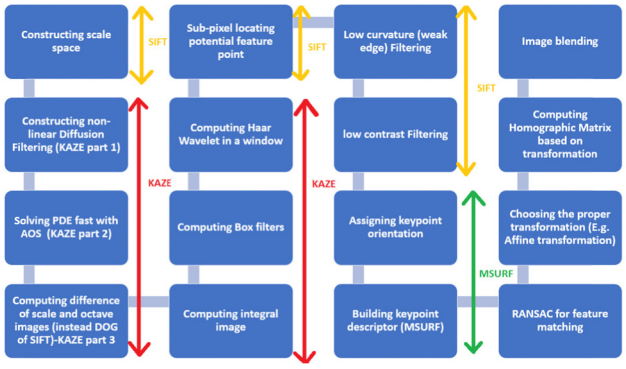


Fig. 2. Block diagram of our algorithm.

capturing, satellites or airplanes have normally a fixed distance to the Earth, and affine transformation can be a proper option for this goal (the trapezius region from rectangular region cannot be shaped). In fact, affine mapping—as a linear transformation—can handle translation, rotation, shearing, and scaling simultaneously.

The block diagram of our method is shown in Fig. 2. The Gaussian filtering step of SIFT is replaced with the nonlinear diffusion filtering step of our method. The difference of images in octaves and scales are replaced with the Haar Wavelet through box filter and integral image. The other parts of the SIFT detector (including subpixel interpolation) have been preserved in our method.

III. Experiments and Analysis

In this section, comprehensive experiments will be conducted to evaluate the accuracy and speed of our algorithm. It is also tested at a different time of imaging, view angles, and source image pairs. Furthermore, it is tested with the presence of noise and distortion. It has also been compared with some well-known image processing descriptors including SIFT, SURF, MSER, BRISK both visually and numerically. All the codes of this section are accessible in GitHub.[1]

All experiments are implemented by MATLAB 2018b in a laptop equipped with an Intel Core i7-7700HG, 2.8-GHz CPU and 16-GB DDR4 RAM with a 64-bit operating system and solid state drive (SSD) Hard.

A. Different Test Images

1) Speckle Noise: Fig. 3(a) and (b) shows two SAR images, which are contaminated with high speckle noise. The SURF method finds some matching features but there are many wrong correspondences so the result of image registration is unsuccessful [Fig. 3(i)]. In fact, the coefficients of affine mapping are inaccurately computed and the first image is incorrectly rotated in z-axes to be stitched to the second image.

The same event takes place for the MSER method [Fig. 3(j)]. The BRISK method has an even worse problem. Although it has found some features [Fig. 3(e) and (f)], only one match is found in the feature matching step, which is lower

[1]For the source code of these experiments please go to the link: https://github.com/Pourfardm/SAR-image-registration-KAZE

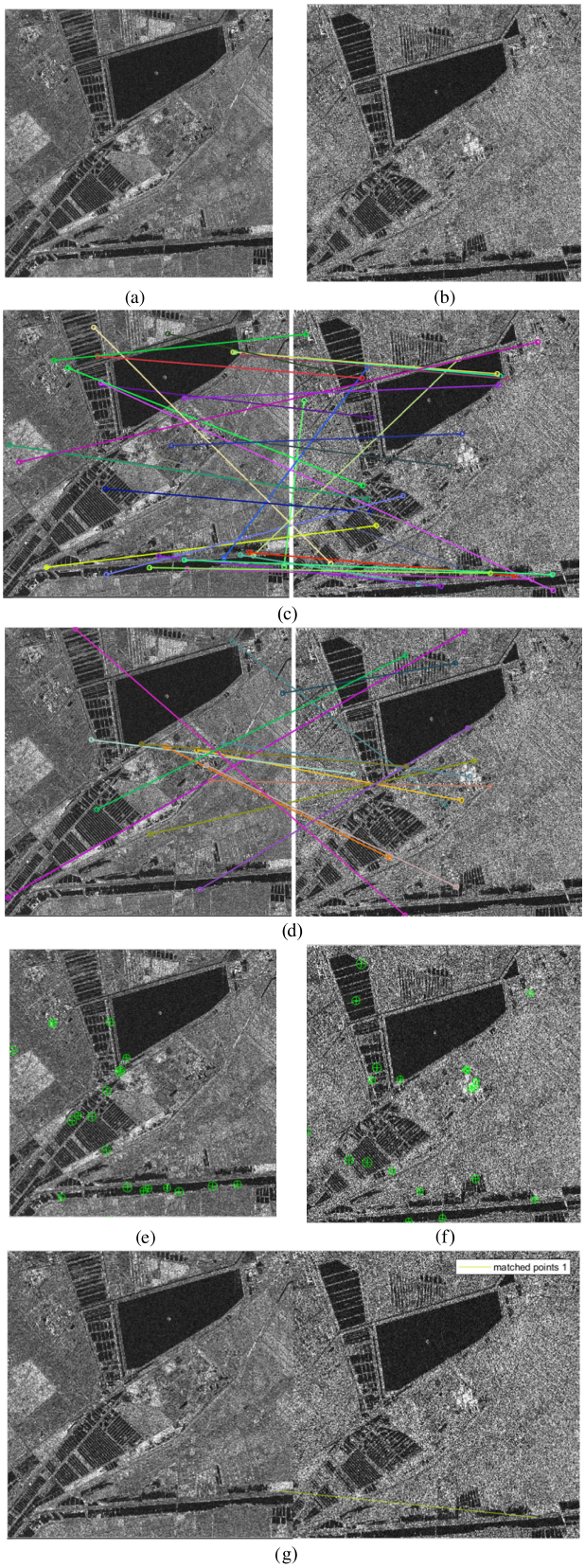


Fig. 3. Result of SURF, MSER, BRISK, and KAZE methods on two SAR images with different view points. (a) First view point [11]. (b) Second view point [11]. (c) Image matching of SURF detector. (d) Image matching of MSER detector. (e) Detected features of (a). (f) Detected features of (b). (g) Image matching of BRISK detector (no complete matching only one match).

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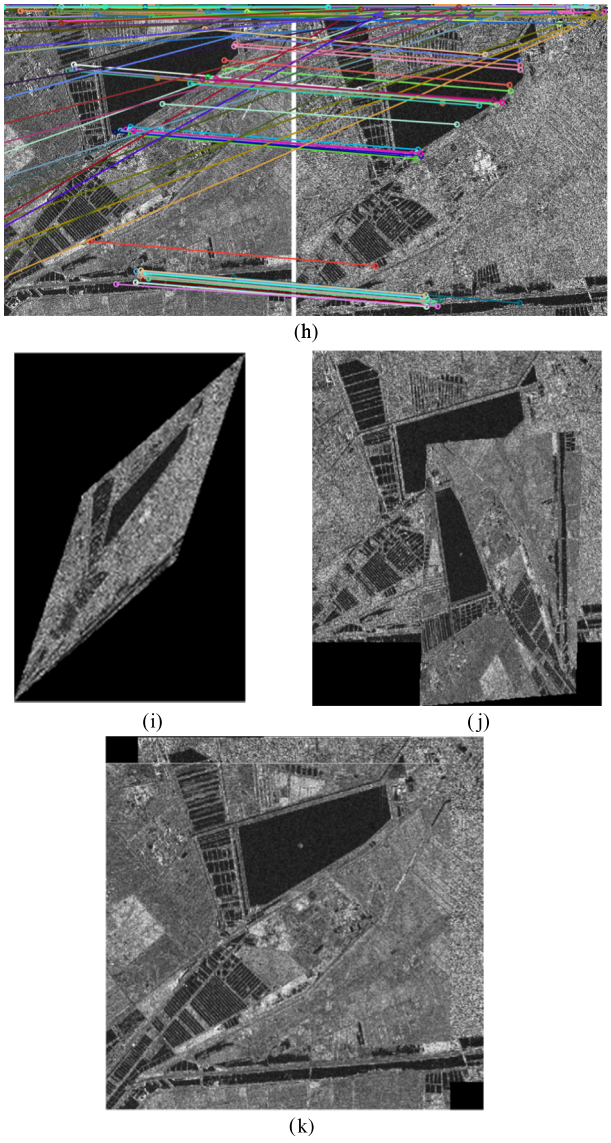


Fig. 3. *(Continued.)* Result of SURF, MSER, BRISK, and KAZE methods on two SAR images with different view points. (h) Image matching of KAZE detector. (i) Image stitching of SURF. (j) Image stitching of MSER. (k) Image stitching of KAZE.

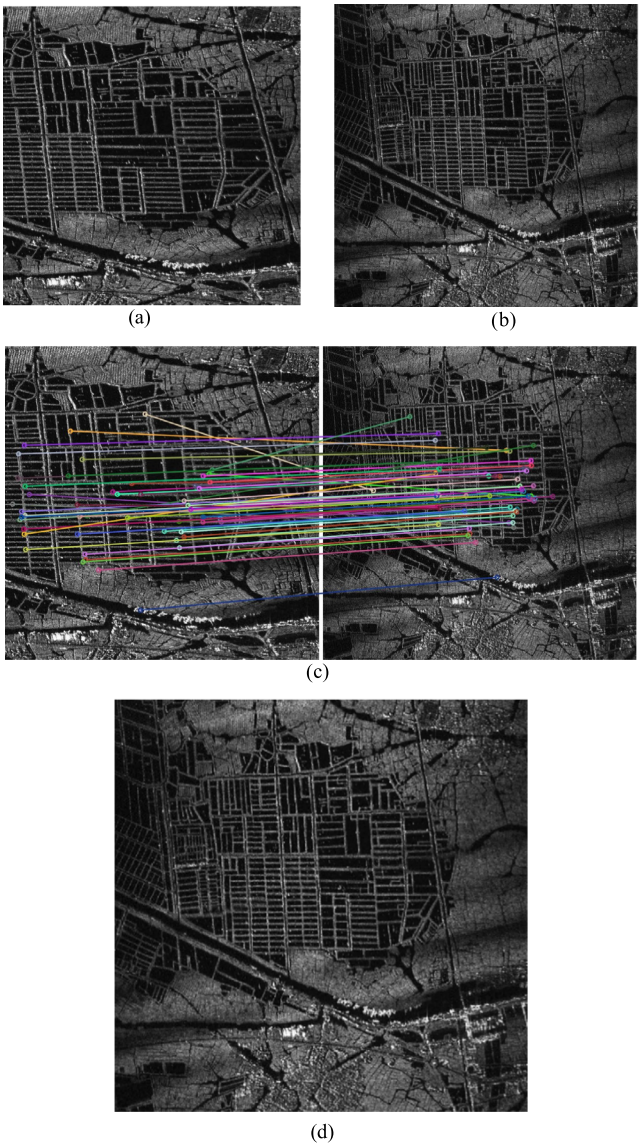


Fig. 4. Result of the KAZE method on two SAR images with different scales in different image sources. (a) Scale 1 [7]. (b) Scale 2 [7]. (c) Image matching. (d) Image stitching.

than the minimum three matches that are needed to compute affine mapping. Meanwhile, KAZE has matched many features correctly [Fig. 3(h)] and has a successful image stitching [Fig. 3(k)]. Although some wrong matches are found due to the presence of speckle noise, the true matches are in majority. This is while all the other methods have found wrong places for image features in the detection step [Fig. 3(c) and (d)]. Therefore, the unsuccessful result of image registration is influenced by the wrong choice of features. Nonlinear diffusion filtering weakens the speckle noise and allows the KAZE method to find true places of image edges.

2) Different Scales: The result of the KAZE method for matching two images with changes in image scales is shown in Fig. 4. As the exact places of image edges are found, the image registration is done without the blurring effect. SURF and MSER are also able to match these two features, while BRISK features are not sufficient for affine mapping.

3) Illumination Change: The illumination of Fig. 5(b) is much more than Fig. 5(a). As a result, the method that is sensitive to an abstract value of gray level such as MSER encounters a problem. The BRISK method is also not able to handle this task. However, the SURF and KAZE correctly register two images. In fact, KAZE is able to simultaneously handle the illumination variation and image affine projection.

4) Different Image Time: Fig. 6 shows three challenging images with speckle noise at different image time.

Although many visual features are visible to be matched with the SURF descriptor, this method is only able to accurately register Fig. 6(a) and (b). In fact, in matching Fig. 6(b) and (c), not only are a few numbers of features detected, but the matching of these features is also done inaccurately.

Therefore, the result of image registration has rotated the third image to match the second image inaccurately [Fig. 6(g)].

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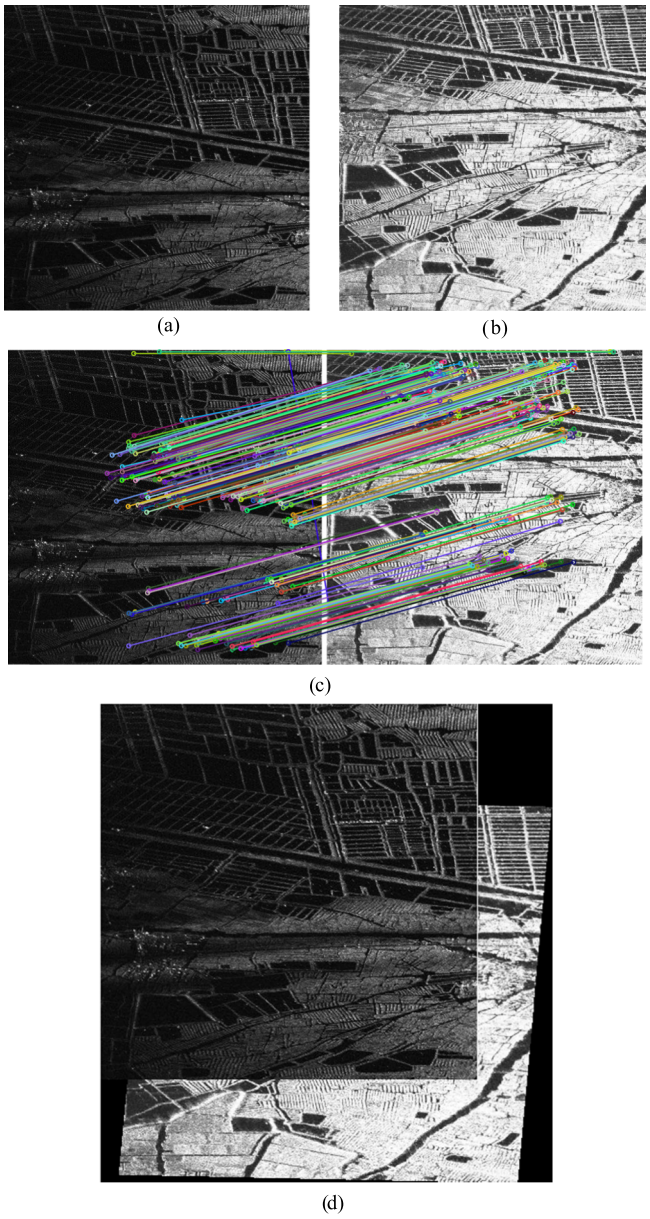


Fig. 5. Result of the KAZE method on two SAR images with different illuminations and affine projection in different image sources. (a) Illumination 1 [7]. (b) Illumination 2 [7]. (c) Image matching. (d) Image stitching.

This is while KAZE has found many more features [Fig. 6(i)] for image matching of Fig. 6(b) and (c) and the final image is constructed truly [Fig. 6(k)].

5) Low Feature Images: Fig. 7 has a smooth structure and there are few details in this image. This location is chosen from Mount Damavand in Iran. This is a challenging task for image registration because of lack of strong edges. Therefore, all the SURF, MSER, and BRISK methods find less than three features, which is insufficient for affine mapping in the image matching step.

Therefore, they are not able to register these images. This is while KAZE finds a few true image features and it also grabs this golden opportunity and matches them correctly.

TABLE I

COMPARISON OF SURF, MSER, BRISK, AND KAZE OF FIGS. 3 AND 5 FOR DIFFERENT CRITERIA

Criterion	Fig 3			
	SURF	MSER	BRISK	KAZE
Image features a	2145	2101	15098	10708
Image features b	3550	4598	25632	12064
Matched features	29	14	1	107
True Positive (TP)	4	3	0	14
Precision	13.80%	21.43%	0	13.08%
Recall	0.07%	0.04%	0	0.06%
F-measure	0.1405	0.0896	0	0.1230
Time (second)	1.10	2.93	-	3.19
Final result	False	False	False	True
Criterion	Fig 5			
	SURF	MSER	BRISK	KAZE
Image features a	1778	1599	5785	11441
Image features b	6021	8784	20482	17670
Matched features	54	5	4	281
True Positive (TP)	20	3	3	106
Precision	37.04%	60%	75%	37.72%
Recall	0.26%	0.03%	0.01%	0.37%
F-measure	0.5129	0.0578	0.0228	0.7282
Time (second)	0.08	4.38	2.08	4.21
Final result	True	False	False	True

Therefore, it is able to register two images correctly. In other words, KAZE reinforces the weak edges through nonlinear diffusion filtering to make them a good feature for image matching.

B. Validation With Precision and Recall

One of the well-known criteria in the image registration method is precision and recall. Precision is the relation of true positive (TP) to all positive values and recall is the relation of TP to all relevant values. Therefore, precision is how useful the search results are and it is a measure of exactness or quality and recall is how complete the results are and is a measure of completeness or quantity.

In image registration, precision quantifies the number of correct matches among all founded matches and recall shows the number of correct matches among all detected features (number of corresponding regions) in two images. In other words, if 50 features are found in sample image A, 40 features are found in sample image B, 20 matches are detected through our feature detection algorithm, and 15 of these feature are correctly visually matched (inlier point), so the precision will be (15/20) and recall will be (15/(50+40-20)). The F-measure encompasses two criteria in one as follows:

$$F = \left(\frac{\text{Precision}^{-1} + \text{recall}^{-1}}{2}\right)^{-1} = 2 \times \frac{\text{Precision} \times \text{recall}}{\text{Precision} + \text{recall}}. \tag{8}$$

The results of our algorithm on Figs. 3 and 5 are shown in Table I. Fig. 3 is a good example due to high speckle noise and Fig. 5 is a good example, which has speckle noise mixed with affine transformation and illumination changes that made it a difficult case for image registration.

The comparison between SURF, MSER, BRISK, and KAZE methods in Fig. 3 shows that both KAZE and BRISK have

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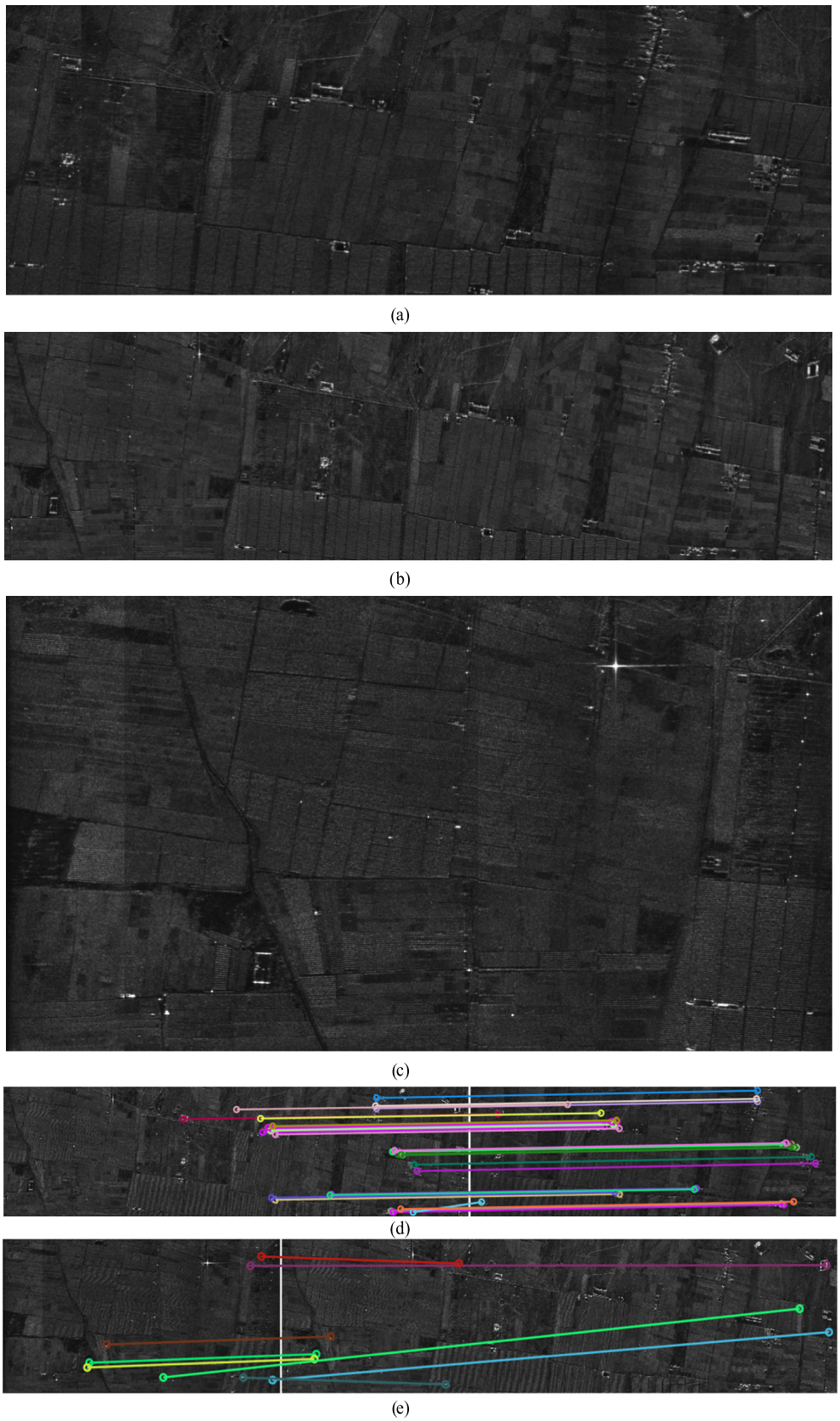


Fig. 6. Comparison of the result of SURF and KAZE detectors on challenging images with speckle noise. (a) Image 1 (2066 × 683). (b) Image 2 (2629 × 6833). (c) Image 3 (1315 × 682). (d) Image matching of (a) and (b) with SURF detector. (e) Image matching of (b) and (c) with SURF detector.

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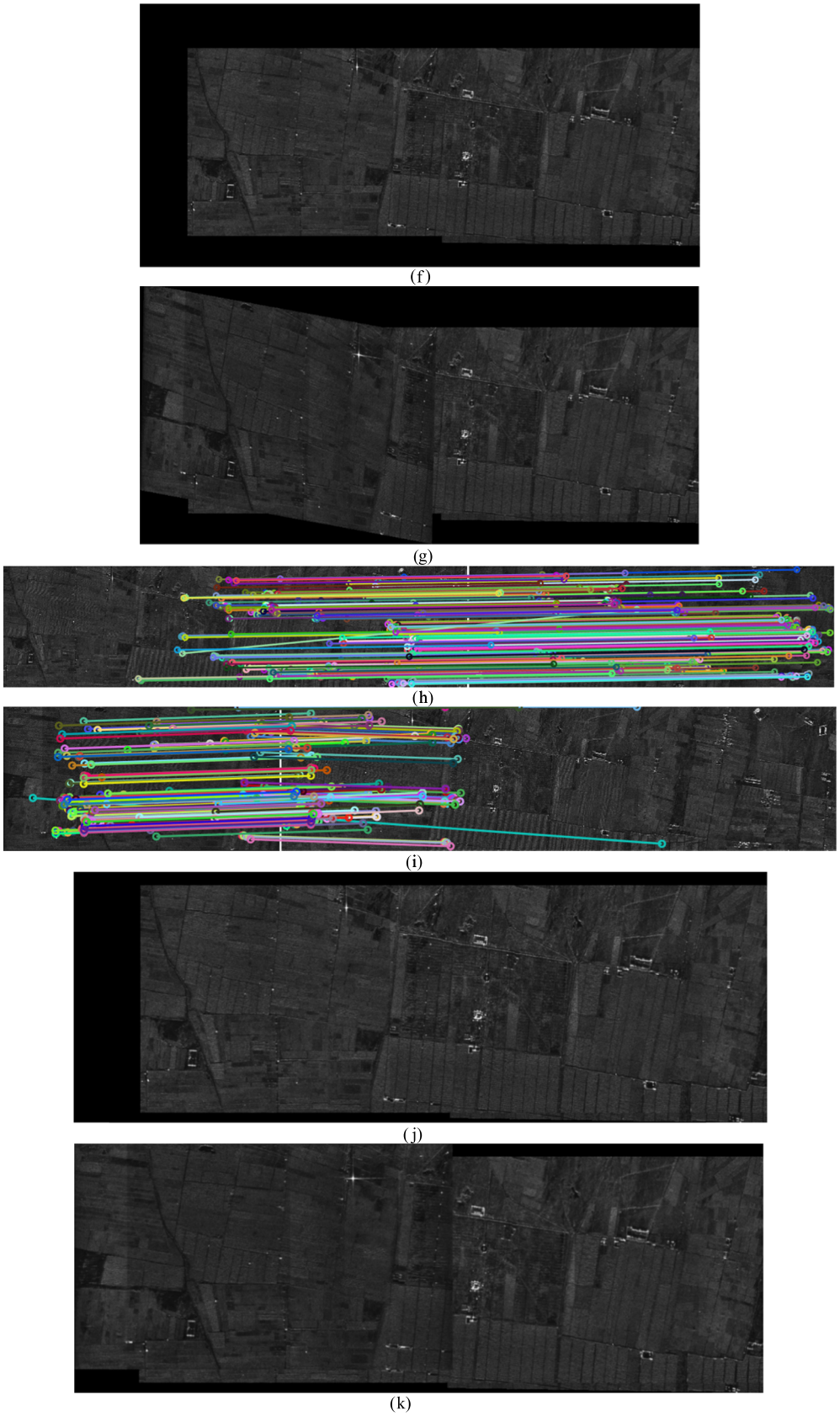


Fig. 6. *(Continued.)* Comparison of the result of SURF and KAZE detectors on challenging images with speckle noise. (f) Image stitching of (a) and (b) with SURF detector. (g) Image stitching of (a)–(c) with SURF detector. (h) Image matching of (a) and (b) with KAZE detector. (i) Image matching of (b) and (c) with KAZE detector. (j) Image stitching of (a) and (b) with KAZE detector. (k) Image stitching of (a)–(c) with KAZE detector.

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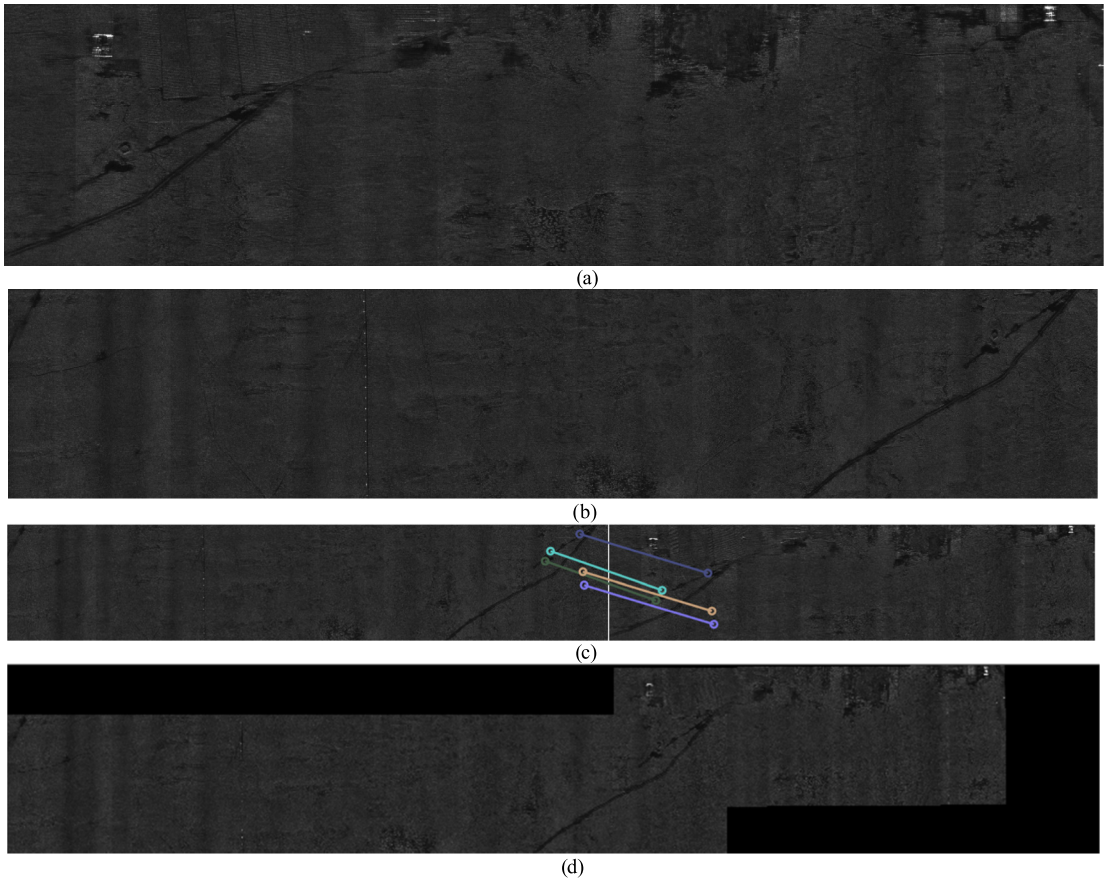


Fig. 7. Result of image registration for the KAZE method in low feature image with speckle noise. (a) Image 1. (b) Image 2. (c) Image matching. (d) Image stitching.

extracted high features in an order of 5 in comparison with MSER and SURF. This is while most BRISK keypoints are not valuable because only 1 match is found for this method and no TP has been found. In fact, high speckle noise has caused this method to be highly sensitive to the absolute value of pixels. KAZE has also selected 107 points nearly three times higher than the SURF method (a faster version of the SIFT method) and all other methods. This method has also selected 14 TP points which is nearly four times higher than all the methods. Based on the precision criteria, MSER has the best result, but this gain is obtained at the cost of low recall among all three methods. It is also noteworthy that the final result of image registration of both MSER and SURF are wrong. In fact, precision only considers the relation of TP to the matched point but it does not guarantee that TP points yield true matches in image registration. Furthermore, although MSER has better precision, it has obtained this result with low matched features (14), while KAZE has obtained this result with high matched features (107). This fact has caused KAZE to have true image registration.

F-measure is a good measure that encompasses both precision and recall. Based on F-measure, KAZE has the best results among them. Therefore, these methods are sorted as KAZE, SURF, MSER, and BRISK based on accuracy. The computational time of KAZE is 3 s while the computation of SURF is 1 s so KAZE has obtained this result at the cost of increasing the computational time in three order of times.

The experiment in Fig. 5 shows that KAZE has obtained the highest number of features even ten times higher than SURF. This method also has the highest number of matched features, which is four times higher than the SURF method. TP points of KAZE are higher than all the other methods.

While KAZE has the third lowest precision compared to BRISK and MSER and a little higher precision than SURF, it is worthy to note that the final result of BRISK and MSER is inaccurate (based on visual judgment). In other words, KAZE has the best result among the successful image registration methods of Fig. 5. Furthermore, KAZE has the best recall among all the methods. The F-measure shows that KAZE is the best method for evaluating accuracy compared to SURF, MSER, and BRISK.

The computational time of MSER in Fig. 5 is the highest because there are lots of gray-scale levels in this image and this increases the computational time of calculating the disjoint areas of this region-based method. Overall, the superiority of accuracy for image registration of KAZE in SAR images is obtained at the cost of increasing the computational time, but it is still of the order of seconds which is an acceptable time.

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In fact, TP can be a good parameter because more detected corresponding features enable the matching step to be more accurate. In other words, the over-determined matrix of affine mapping can be solved more accurately by having more equations than unknowns [33]. The second parameter is F-measure. It encompasses both precision and recall in itself. The third parameter is the final result (true or false). If we have a good F-measure but a wrong result, it does not have any value. This can be evaluated visually or by a ground-truth image. Therefore, the multiplication of F-measure and final result (one or zero) is introduced as our KAZE-SAR accuracy evaluation parameter.

It is also noteworthy that if the modified SURF descriptor is replaced with the SIFT while preserving the KAZE detector, the accuracy of the result does not change (e.g., the true matching of Fig. 3 will be repeated). However, the computational cost of the algorithm increases due to double dimensions of SIFT.

On the other hand, if the KAZE detector is replaced with the SIFT detector while preserving the modified SURF descriptor, the accuracy of the result decreases (e.g., the matching of Fig. 3 will be wrong similar to SURF detector). However, the computational cost will be nearly at in the same order. Although nonlinear diffusion filtering as a time-consuming step is added to our method, we have used the integral image and Haar wavelet steps in our method instead of Laplacian filtering steps of SIFT, which are faster and compensate the other part.

IV. Conclusion

In this study, for the first time, the KAZE method as a detector is mixed with modified version of SURF descriptor for image registration of the SAR images. Speckle noise is nonlinear noise, which is created in image acquisition of the SAR image; therefore, the normal image denoising methods are not able to completely eliminate this noise. This noise can have a direct influence on image matching and image stitching. Our method uses nonlinear diffusion filtering to decrease the effect of this noise and prevent our feature detector model from getting trapped in the local minimum of speckle noise. This is done by solving discrete stochastic second-order nonlinear PDEs.

An innovation in mathematics has enabled us to solve this equation at such a fast pace and allowed us to use this method for SAR image registration. Extensive experimental results confirm our claim both visually and numerically based on the absolute value of TP and F-measure criterion. This achievement is gained at the cost of higher computational cost. For future works, the mixture of SAR detector with deep learning can be used to increase the accuracy of image matching.

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